

Homework 7 Solutions

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Problem 15.1

- a) 2,5,10
- b) 5
- c) 2,3,6,9,18,27,54
- d) Nothing.

Problem 15.2

Proof. If $a, b \in \mathbb{Z}$, then the largest integer N such that $a \equiv b \pmod{N}$ is $|a - b|$.

- 1. Let $a, b \in \mathbb{Z}$ such that $a \equiv b \pmod{N}$.
- 2. Since $a \equiv b \pmod{N}$ then $N|(a - b)$, and so $\exists k \in \mathbb{Z}$ such that $Nk = (a - b)$.
- 3. Observe that N reaches its max when $k = 1$ or $k = -1$ depending on if $(a - b)$ is positive or negative.
- 4. In either case, N achieves a max of $N = |a - b|$. □

Problem 15.3

- a) Equivalence relation.
- b) Not an equivalence relation.
- c) Not an equivalence relation.
- d) Not an equivalence relation.
- e) Equivalence relation.
- f) Not an equivalence relation.
- g) Equivalence relation.

Problem 15.4

Proof. If x and y are both odd integers, then $x \equiv y \pmod{2}$.

- 1. Let x and y be odd integers.
- 2. Then $x = 2k_0 + 1$, and $y = 2k_1 + 1$. (Definition of odd)
- 3. Then $x - y = (2k_0 + 1) - (2k_1 + 1) = 2(k_0 - k_1)$.
- 4. Since $\exists k \in \mathbb{Z}$ such that $2k = (x - y)$, namely $k = k_0 - k_1$, then $2|(x - y)$. (Definition of divisible)
- 5. Since $2|(x - y)$, then $x \equiv y \pmod{2}$. (Definition of congruence modulo)
- 6. Therefore if x and y are both odd integers, then $x \equiv y \pmod{2}$. □

Proof. If x and y are both even integers, then $x \equiv y \pmod{2}$.

- 1. Let x and y be even integers.
- 2. Then $x = 2k_0$, and $y = 2k_1$. (Definition of even)
- 3. Then $x - y = 2k_0 - 2k_1 = 2(k_0 - k_1)$.

4. Since $\exists k \in \mathbb{Z}$ such that $2k = (x - y)$, namely $k = k_0 - k_1$, then $2|(x - y)$. (Definition of divisible)
5. Since $2|(x - y)$, then $x \equiv y \pmod{2}$. (Definition of congruence modulo)
6. Therefore if x and y are both even integers, then $x \equiv y \pmod{2}$. □

Problem 15.5

Proof. If $a \in \mathbb{Z}$, then $a \equiv -a \pmod{2}$.

1. Let $a \in \mathbb{Z}$.
2. Then $a - (-a) = 2a$.
4. Since $\exists k \in \mathbb{Z}$ such that $2k = (a - (-a))$, namely $k = a$, then $2|(a - (-a))$. (Definition of divisible)
5. Since $2|(a - (-a))$, then $a \equiv -a \pmod{2}$. (Definition of congruence modulo)
6. Therefore if $a \in \mathbb{Z}$, then $a \equiv -a \pmod{2}$. □

Problem 15.6

Proof. Congruence modulo n is transitive.

1. Let $x \equiv y \pmod{n}$, and $y \equiv z \pmod{n}$.
2. Then $n|x - y$ and $n|y - z$. (Definition of congruence mod)
3. Then $x - y = nk_0$ and $y - z = nk_1$. (Definition of divisible)
4. Adding the equations from 3 results in $x - z = n(k_0 + k_1)$.
5. Since $\exists k \in \mathbb{Z}$ such that $nk = x - z$, namely $k = k_0 + k_1$, then $n|(x - z)$. (Definition of divisible)
6. Since $n|(x - z)$, then $x \equiv z \pmod{n}$. (Definition of congruence mod)
7. Therefore congruence modulo n is transitive. □

Problem 15.7

- a) $\{1, 2\}$
- b) $\{4\}$
- c) $\{120, 121, 122, 123, 124, 125, 126, 127, 128, 129\}$
- d) The set containing my siblings.
- e) The set of all people born on November 10th.
- f) $\{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \{3, 4\}, \{3, 5\}, \{4, 5\}\}$

Problem 15.12

Proof. If R is an equivalence relation on set A such that $x, y \in [a]$, then xRy .

1. Let R be an equivalence relation on set A such that $x, y \in [a]$.
2. Then xRa and yRa .
3. Since R is symmetric and yRa then aRy , and then since R is transitive, xRa , and aRy then xRy .
4. Therefore if R is an equivalence relation on set A such that $x, y \in [a]$, then xRy . □

Problem 16.1

1. $\{\{1, 2, 3\}\}, \{\{1, 2\}, \{3\}\}, \{\{1, 3\}, \{2\}\}, \{\{2, 3\}, \{1\}\}, \{\{1\}, \{3\}, \{2\}\}$
2. $\{\{1, 2, 3, 4\}\}, \{\{1, 2, 3\}, \{4\}\}, \{\{1, 2, 4\}, \{3\}\}, \{\{1, 3, 4\}, \{2\}\}, \{\{2, 3, 4\}, \{1\}\}, \{\{1, 2\}, \{3, 4\}\}, \{\{1, 3\}, \{2, 4\}\}, \{\{1, 4\}, \{2, 3\}\}, \{\{1\}, \{2\}, \{3\}, \{4\}\}$

Problem 16.2

- a) $6!$
- b) $\frac{8!}{2!}$
- c) $\frac{11!}{2!2!2!}$
- d) $\frac{7!}{3!2!}$
- e) $\frac{11!}{4!4!2!}$

Problem 16.3

$$\frac{5!}{2!}$$

Problem 16.4

$$6! \cdot \binom{11}{5}$$

Problem 16.5

Proof. Proposition 16.4.

1. Let $[x]$ be an equivalence class on is-in-the-same-part-as relation, and let $y \in [x]$.
2. Since $x, y \in [x]$, then xRy .
3. Since xRy , then $\exists$ a part P such that $x, y \in P$.
4. Therefore $[x] \subseteq P$.
5. Let P be some part, and let $x, y \in P$.
6. Since x and y are in the same part, then xRy .
7. Since xRy , then $x, y \in [x]$.
8. Therefore $P \subseteq [x]$.
9. Therefore $P = [x]$.

□

Problem 16.7

$$11!$$

Problem 16.15

For $\{1, 2, 3, 4\}$, the answer is 7.

For $\{1, 2, 3, \dots, 100\}$ the answer is $2^9 - 1$.

In general, $2^{n-1} - 1$ where n is the size of the set.